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Pages: 2

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Eighth Semester B.Tech Degree Regular Examination June 2023 (2019 Scheme)

Course Code: EET468

Course Name: INDUSTRIAL INSTRUMENTATION AND AUTOMATION

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|--|-----|
| 1 | What is meant by sensor-time response? | (3) |
| 2 | List the various factors influencing the choice of a transducer | (3) |
| 3 | Explain the operation of a charge amplifier | (3) |
| 4 | Explain the principle and operation of phase sensitive detector. | (3) |
| 5 | Explain Process control network | (3) |
| 6 | List the advantages of virtual instrumentation systems | (3) |
| 7 | Give the significance of timers and counters in PLC | (3) |
| 8 | What is ladder programming | (3) |
| 9 | Explain the role of MTU in SCADA communication | (3) |
| 10 | Explain the concept of Human Machine Interface in DCS | (3) |

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

- | | | |
|----|--|-----|
| 11 | a) With necessary diagram and equations explain about second order response of a sensor. | (8) |
| | b) With necessary diagrams, explain self-regulation, human aided control and automatic control in a process. | (6) |

OR

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|----|--|-----|
| 12 | a) Illustrate Capacitive differential pressure measurement | (7) |
| | b) Explain displacement measurement using Resistance Potentiometer | (7) |

Module II

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| 13 | a) Derive an expression for the output voltage of a logarithmic amplifier and show that it is proportional to the logarithm of input voltage. | (8) |
| | b) Explain Differential amplifier | (6) |

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- 14 a) Describe various Electrical Actuators (8)
b) Write short notes on Fluid valves (6)

Module III

- 15 a) Explain about Fiber optic data transmission (8)
b) Compare radio and wireless communication (6)

OR

- 16 a) Explain Pneumatic transmission (8)
b) Compare cable transmission analog and digital types (6)

Module IV

- 17 a) Explain with an example the working of counters in PLC. (8)
b) Explain the importance of power supply in PLC (6)

OR

- 18 a) Draw the ladder diagram for realising NAND & XOR logic using PLC. (8)
b) Explain the importance of CPU in PLC (6)

Module V

- 19 a) Explain the architecture of DCS. (8)
b) What are the main components of SCADA (6)

OR

- 20 a) Explain the advantages, limitations and applications of DCS (8)
b) Explain the significance and features of the DNP3 protocols used in SCADA systems. (6)
